Kalam smiled.

'It's an abbreviation for Dreamer1,' he said. 'It may remain a dream; it may not take off at all.'

Kalam was right. This dreamer never took off.

Though Gowarikar saw me as a Muthunayagam man, I was not interested in their politics, much less take sides. My focus was on getting the liquid propulsion Vikas engine into the PSLV project, which was now being configured on paper.

Any rocket is configured top-down. That is, you first decide on the mass of the payload and the orbit it has to be launched into; then you work out permutations and combinations of the stages of the rocket that would take the mass to the desired altitude. Teams would do mathematical calculations and simulations with various configurations to arrive at the best option.

The same was being done simultaneously for PSLV and GSLV. S Srinivasan headed the PSLV planning group, Ratnaraj Jayamani was in charge of GSLV planning. GSLV, which needed a cryogenic engine appeared a less immediate reality than PSLV, and I saw in PSLV the right opportunity to incorporate the liquid Vikas engine.

By virtue of being the deputy director of the propulsion group, I participated in the PSLV planning group presentations and discussions. The group worked out hundreds of configurations, toying with different ideas from a single-stage rocket to a four-stage rocket in different combinations of solid propulsion engines. Not one of the configurations had a liquid engine in it.

Whenever I spoke of trying out a liquid stage, there were either smirks or silence. But I did not give up. I would present scientific facts that showed the supremacy of liquid engines over solid, like the clean cut-off of the stage after burning. A liquid engine can burn for a much longer duration than a solid engine; a liquid engine can be cut off and re-ignited, while a solid engine once ignited has no control over, and has to burn out on its own.

Then there is the case of energy levels, called specific impulse in rocket science. A solid propulsion engine has a maximum specific impulse of 240